

Module 3:

Digital Forensics



Topic 2: Steganography



What is Steganography?

- The art of hiding secret data within unrelated data or medium.
- The difficulty with steganography is that extracting the hidden message requires not only a detection that steganography has been used, but also the [exact stenographic](#) tool used to embed it.



Steganography VS Cryptography

The target of both is to protect a message or piece of information

But the difference:

- **Cryptography:** Changes message from readable format to unreadable format, only the authorized person will have the key to revert it back to readable format.
- **Steganography:** Hides existence of the message itself so only the right recipient will know where it's hidden and how to reveal it.

Steganography Techniques

1. Image Steganography
2. File Steganography
3. Audio Steganography

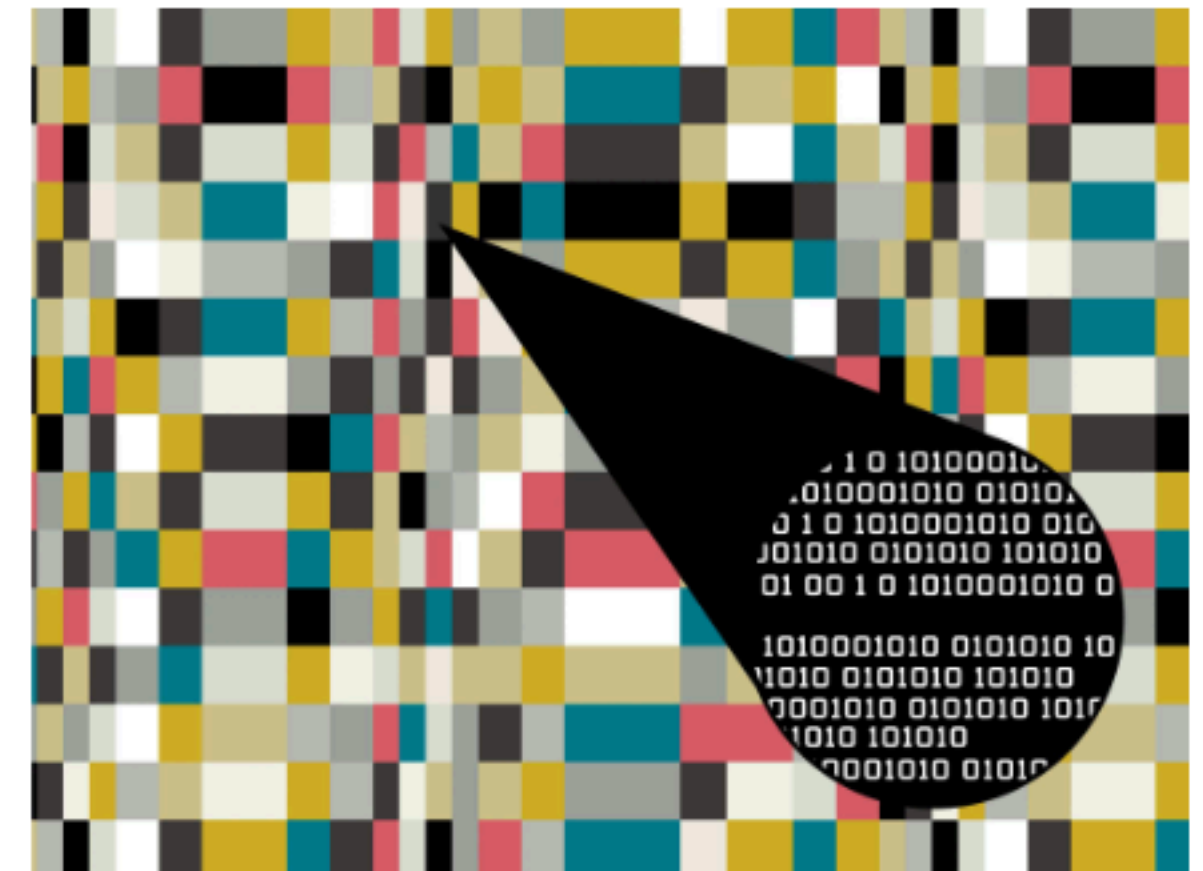
Steganography Techniques

Image Steganography:

Common method in which information is hidden inside an image in the way that will not be detectable except for the recipient itself

Some techniques:

- Least significant bit insertion
- Masking and filtering



Lab

Images3c



Steganography Techniques

Audio Steganography

The secret message is hidden inside audio file, this is more challenging technique because the ear can detect the changes more than eyes. The challenge is how to hide the message without being detected

Common techniques:

- Least Significant bit Coding
- Parity coding

One of the easiest techniques is hiding the message in the frequencies above 20 KHZ and below 20 HZ

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I Love Music



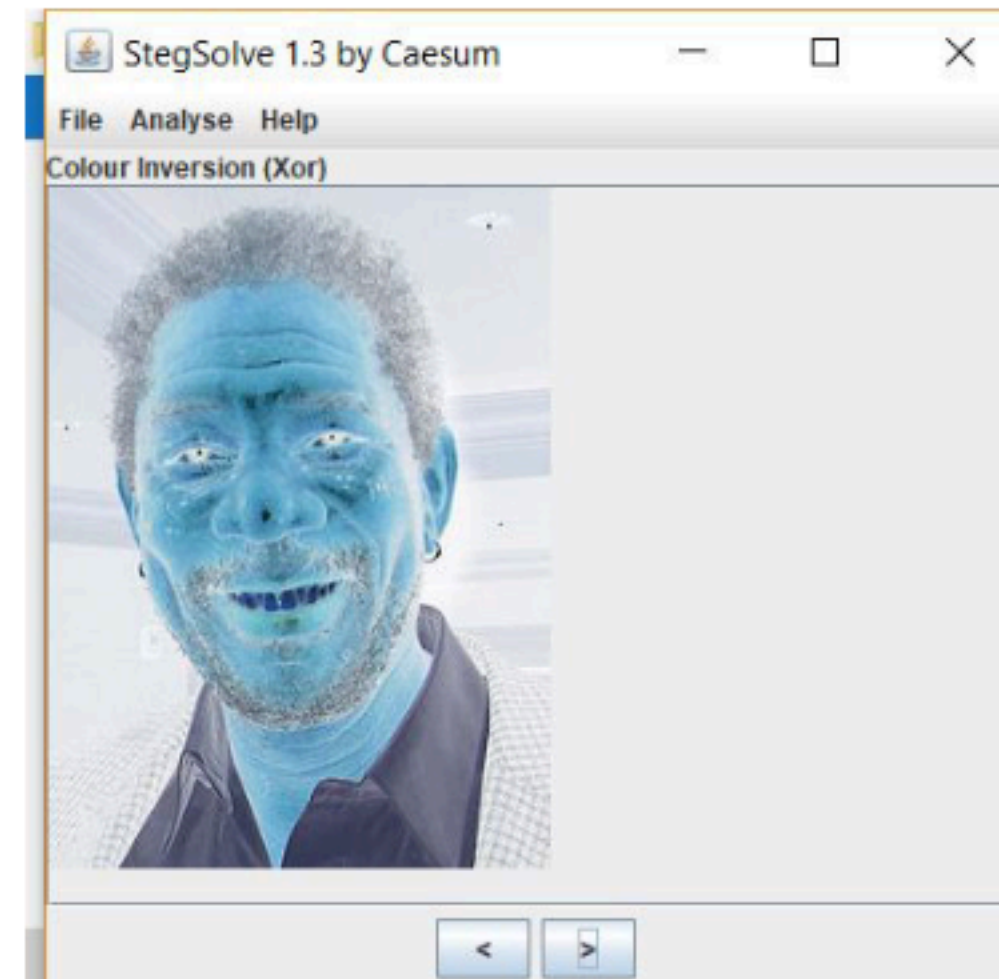
Steganography Techniques

File Steganography

Old technique of hiding images or text files inside other files. The original file will not be impacted however this can be detected by size detection or by forensics/analysis tools

Best Tools to Perform Steganography

Stegsolve: is often used to apply various steganography techniques to image files in an attempt to detect and extract hidden data.



Best Tools to Perform Steganography

Photoshop or GIMP: is used to view the hue / saturation levels of any image.

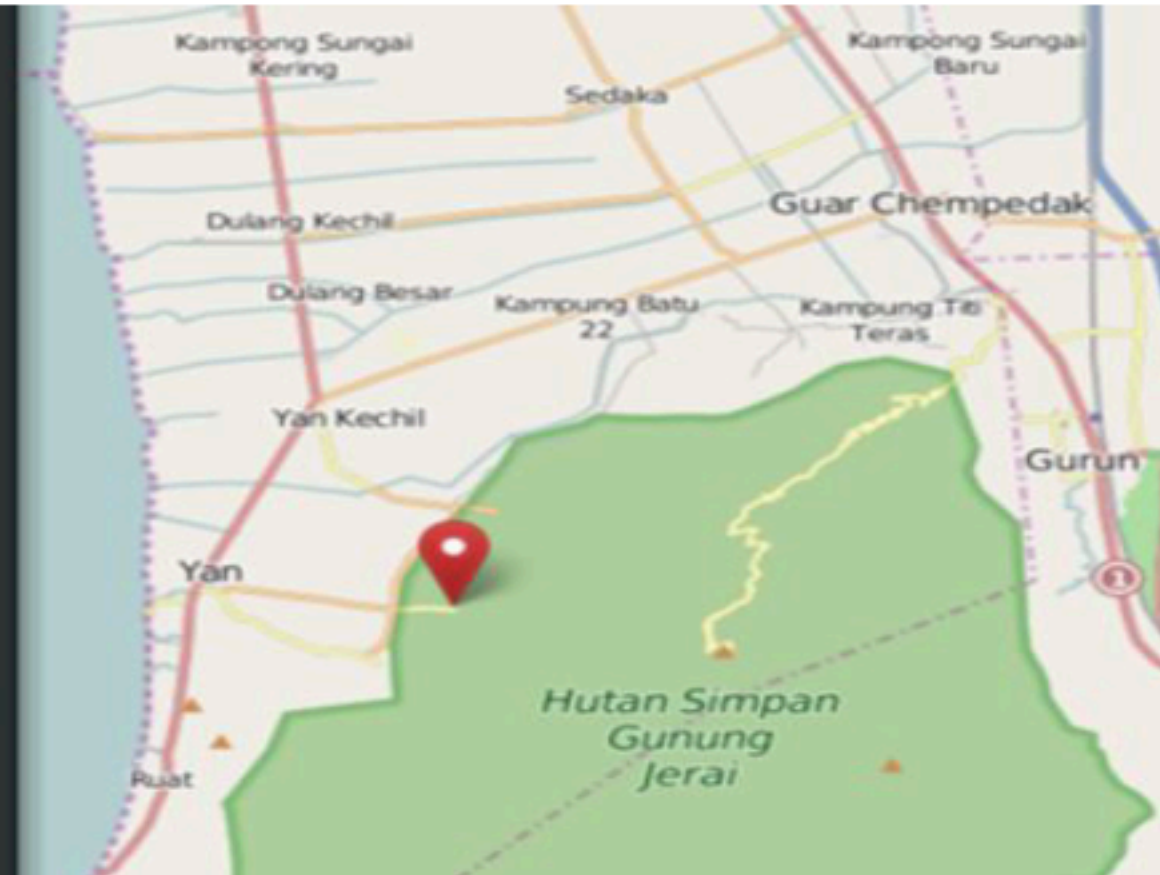


Best Tools to Perform Steganography

Important tools that gives more information about an image -
Dimensions, Location: **exiftool**

GPS location Mapping:

```
GPS Latitude Ref : North
GPS Longitude Ref : East
GPS Altitude Ref : Above Sea Level
GPS Time Stamp : 11:07:47
GPS Img Direction Ref : True North
GPS Img Direction : 82.12307692
GPS Date Stamp : 2011:09:04
XMP Toolkit : XMP Core 5.1.2
Creator Tool : 4.3.5
Date Created : 2011:09:04 12:51:11
Image Width : 1024
Image Height : 765
Encoding Process : Baseline DCT, Huffman coding
Bits Per Sample : 8
Color Components : 3
Y Cb Cr Sub Sampling : YCbCr4:2:0 (2 2)
Aperture : 2.8
GPS Altitude : 0 m Above Sea Level
GPS Date/Time : 2011:09:04 11:07:47Z
GPS Latitude : 38 deg 54' 35.40" N
GPS Longitude : 1 deg 26' 19.20" E
GPS Position : 38 deg 54' 35.40" N, 1 deg 26' 19.20" E
Image Size : 1024x765
Megapixels : 0.783
Shutter Speed : 1/3016
Focal Length : 3.9 mm
Light Value : 14.9
```



Best Tools to Perform Steganography

Steghide: also from the most famous tools in the field of Steganography. It may require a password to extract a file.

```
steghide version 0.5.1
the first argument must be one of the following:
embed, --embed          embed data
extract, --extract      extract data
info, --info            display information about a cover- or stego-file
info <filename>        display information about <filename>
encinfo, --encinfo      display a list of supported encryption algorithms
version, --version      display version information
license, --license      display steghide's license
help, --help            display this usage information

embedding options:
-ef, --embedfile         select file to be embedded
-ef <filename>          embed the file <filename>
-cf, --coverfile         select cover-file
-cf <filename>          embed into the file <filename>
-p, --passphrase         specify passphrase
-p <passphrase>         use <passphrase> to embed data
-sf, --stegofile         select stego file
-sf <filename>          write result to <filename> instead of cover-file
-e, --encryption         select encryption parameters
-e <a>[<m>][<m>[<a>]    specify an encryption algorithm and/or mode
-e none                 do not encrypt data before embedding
-z, --compress           compress data before embedding (default)
-z <l>                  using level <l> (1 best speed...9 best compression)
-Z, --dontcompress      do not compress data before embedding
-K, --nochecksum         do not embed crc32 checksum of embedded data
-N, --dontembedname     do not embed the name of the original file
-f, --force             overwrite existing files
-q, --quiet             suppress information messages
-v, --verbose           display detailed information

extracting options:
-sf, --stegofile         select stego file
-sf <filename>          extract data from <filename>
-p, --passphrase         specify passphrase
-p <passphrase>         use <passphrase> to extract data
-xf, --extractfile       select file name for extracted data
-xf <filename>          write the extracted data to <filename>
-f, --force             overwrite existing files
-q, --quiet             suppress information messages
-v, --verbose           display detailed information

options for the info command:
-p, --passphrase         specify passphrase
-p <passphrase>         use <passphrase> to get info about embedded data

To embed emb.txt in cvr.jpg: steghide embed -cf cvr.jpg -ef emb.txt
To extract embedded data from stg.jpg: steghide extract -sf stg.jpg
```

Best Tools to Perform Steganography

binwalk: is great for checking out if other files are embedded or appended to a file.

sudo apt-get install binwalk

```
root@kali:~/Desktop# binwalk -e widescreen.png
  widescreen.png
DECIMAL      HEXADECIMAL      DESCRIPTION
-----
0            0x0              PNG image, 628 x 281, 8-bit/color RGB, non-interlaced
85           0x55              Zlib compressed data, best compression
2757         0xAC5             Zlib compressed data, best compression

root@kali:~/Desktop# cd _widescreen.png.extracted/
root@kali:~/Desktop/_widescreen.png.extracted# ls
55  55.zlib  AC5  AC5.zlib
root@kali:~/Desktop/_widescreen.png.extracted#
```

Best Tools to Perform Steganography

Foremost is a forensic data recovery program for Linux used to recover files using their headers, footers, and data structures through a process known as file carving.

```
sudo apt-get install foremost
```

```
sudo apt install ncurses-hexedit
```



Best Tools to Perform Steganography

One of the famous tricks of Steganography is to hide information in an audio or video file.

Tools to use:

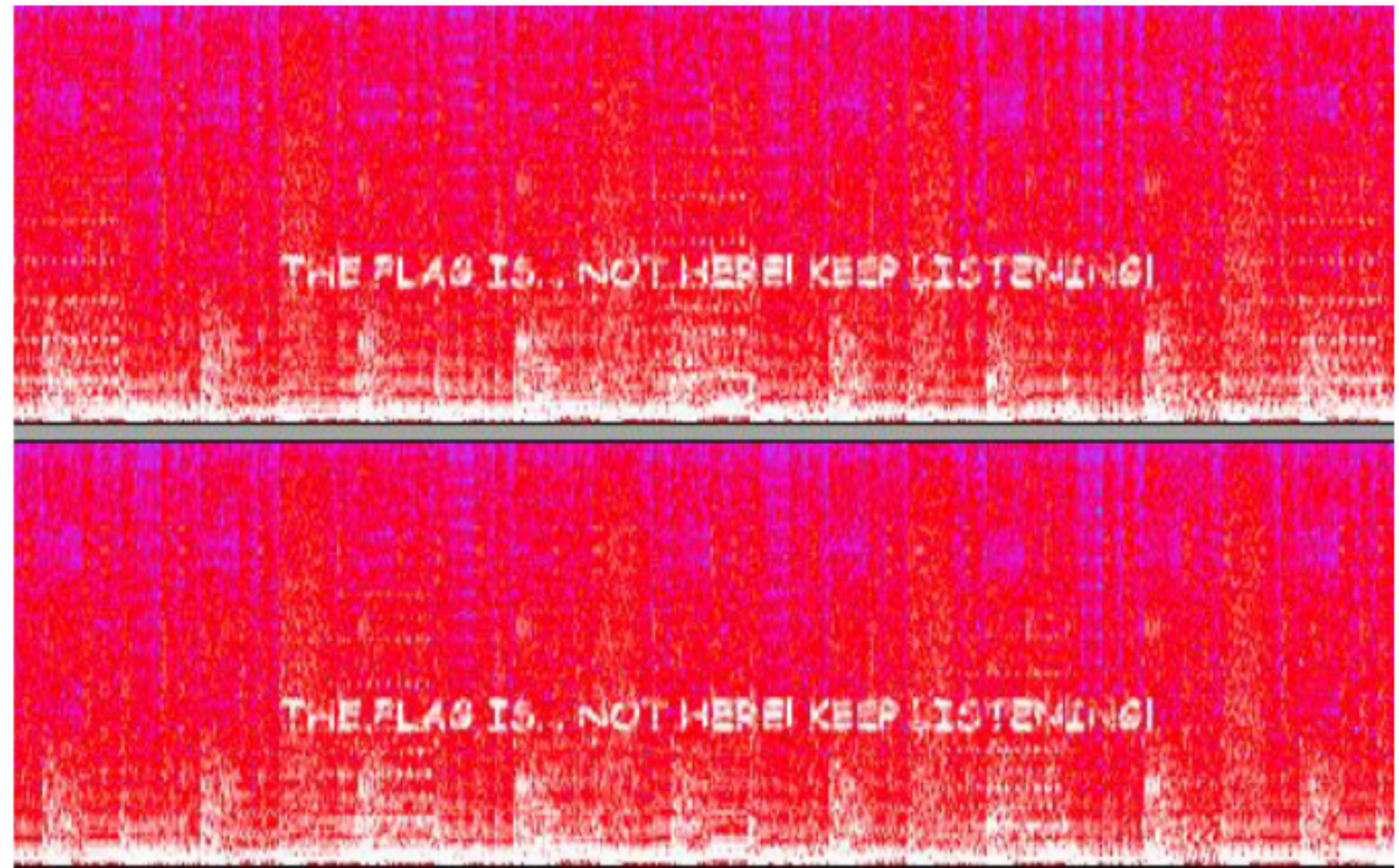
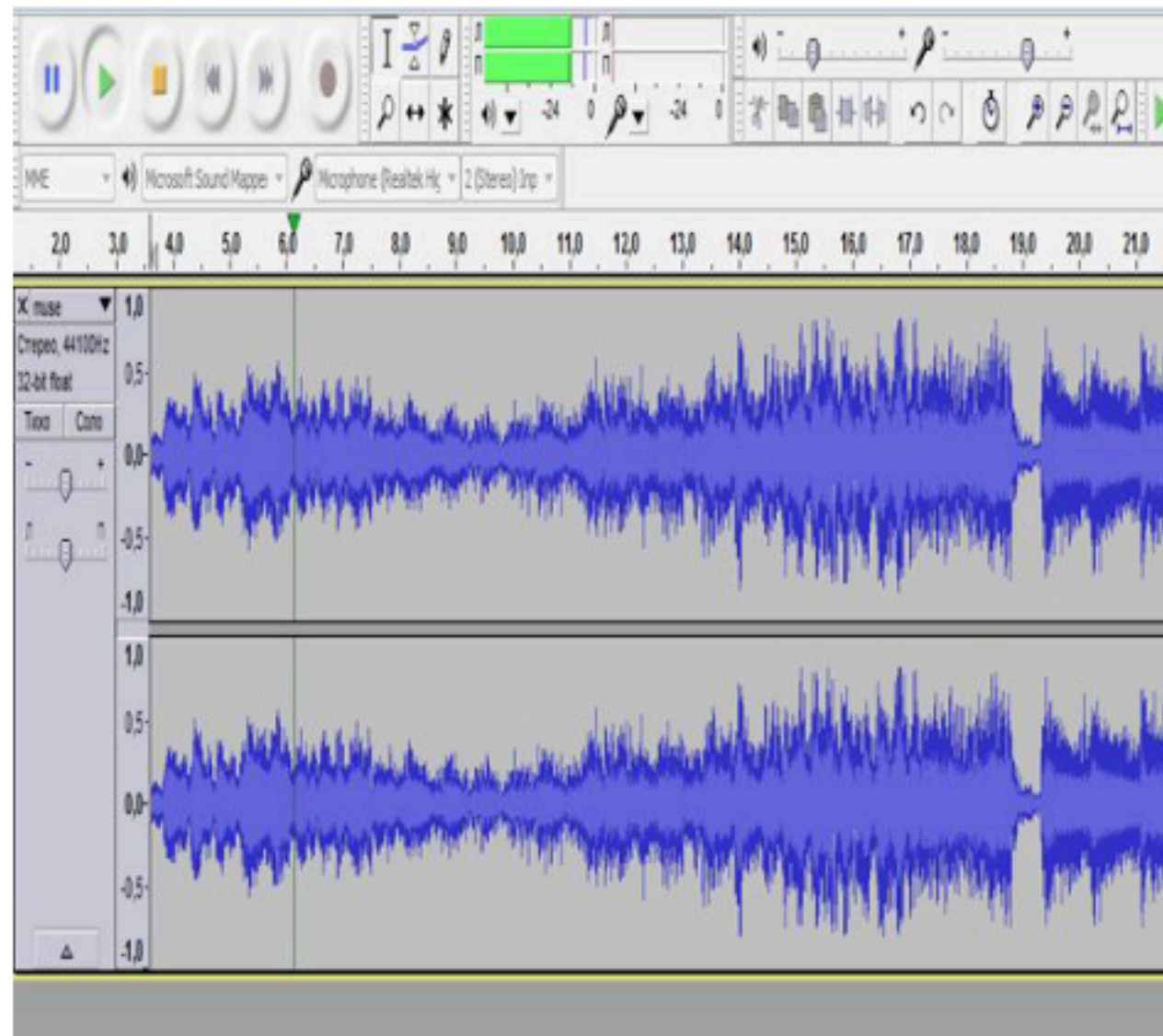
Audacity: open-source audio file and waveform-viewing tool, and it can be used to extract text from audio waveforms

`sudo apt-get install audacity`

Deep Sound: <https://jpinsoft.net/deepsound>

Best Tools to Perform Steganography

Audacity



Best Tools to Perform Steganography

Deep Sound

